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Applying Kansei Engineering and data mining to design door-to-door delivery service

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ABSTRACT

This study proposes a service design approach integrating Kansei Engineering and a data mining technique, in which Kansei Engineering is an ergonomic approach of customer-oriented product/service development and can translate the users' subjective perceptions into a design specifications. The integrated approach collects customers' relevant perceptual vocabulary and service properties based on the Kansei Engineering procedure. Subsequently, it quantifies the relationship among service properties, perceptual responses and usage intention through the data mining technique using a decision tree. A case of door-to-door delivery (D2DD) service is adopted to demonstrate that the proposed approach can incorporate the customers' feelings into the process of service design or improvement and illustrate that the decision tree is suitable to be integrated with Kansei Engineering. The analytical results show the influence of a combination of different service properties (resp. perceptual responses) on a perceptual response (resp. usage intention). It is found that the combinations of crucial perceptual responses result in positive (resp. negative) usage intention and the property combinations result in these crucial perceptual responses. Accordingly, the D2DD service provider can improve or create its service based on the research findings.

1. Introduction

From the perspective of service management, customers always expect to gain 'a great vibe' from services. However, the feeling each customer gets from using services may not be immutable. As a customer becomes habituated to a certain quality level of service, any initial delight may subside over time, and the customer may simply expect that service going forward (Kano, Seraku, Takahashi, & Tsuji, 1984). Thus, services should be improved or innovated continuously to meet the changing requirements and perceptions of customers (Fitzsimmons & Fitzsimmons, 2006).

Kansei Engineering proposed by Nagamachi (1989) is an ergonomic approach of customer-oriented product development to relate specific customer feelings to attributes of the product and associated improvements. Kansei Engineering is different from quality function development (QFD) and conjoint analysis that deal with customers' explicit needs (Cohen, 1995; Green & Srinivasan, 1990), and it proposes to measure customers' implicit needs and to associate them with the product properties (Lévy, 2013). Since service is intangible and mainly utilized to satisfy the customers' perceived requirements, it is considerable to adopt Kansei Engineering to translate the feelings they get from using the service for service design and improvement. According to Nagamachi (1989, 1995) and Demirtas, Anagun, and Koksal (2009), Kansei Engineering adopts perceptual vocabulary to describe the customers' feelings about a specific product rather than using technical terms. Most importantly, Kansei Engineering seeks to determine the cause-and-effect relationship between perceptual responses and design properties to develop a quantitative model (Yang, 2011). So far, Kansei Engineering has been applied to many kinds of product designs, such as automotive interiors (Nagamachi, 1995), website interfaces (Lokman & Nagamachi, 2009), mobile phones (Shi, Sun, & Xu, 2012; Yang, 2011), and architecture (Llinares & Page, 2011), but to our knowledge only Chen, Chang, Hsu, and Xiao (2015) and Chen, Hsu, Chang, and Chou (2015) utilized Kansei Engineering for service design.

Kansei Engineering can, therefore, be viewed as a framework combining one of several analytical tools including Semantic Differential Analysis, Principal Component Factor Analysis, Logistic Regression, and Partial Least Squares, to build the cause-and-effect relationship between perceptual responses and design properties (Llinares & Page, 2011; Schütte, Eklund, Axelsson, & Nagamachi, 2004). For example, Chen, Chang, et al. (2015) and Chen, Hsu, et al. (2015) utilized Logistic Regression and Partial Least Squares,

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respectively, to analyze the cause-and-effect relationship between perceptual responses and service properties. However, Gustafsson and Johnson (2004) pointed out that both Logistic Regression and Partial Least Squares ignore some effects from the combinations of independent variables. To take into account such effects, this study integrates Kansei Engineering and a data mining technique, Decision Tree, to develop several recommendations for describing the relationship between service properties and customers' perceptual responses.

Consider that a well-designed and high-quality door-to-door delivery (D2DD) service can promote the online transaction market and affect customers' usage intentions (Cairns, 1996; Kwek, Tan, & Lau, 2010; Chen, Chang, Hsu, & Yang, 2011; Liao, Chen, & Lin, 2011). Therefore, a case of D2DD service design is adopted to demonstrate the usability of the proposed approach integrating Kansei Engineering and a decision tree and show the performance of the decision tree by comparing it with a logistic regression. At first, the perceptual vocabulary words related to usage experience of D2DD are collected and induced via interviews, websites, magazines, and papers. Such a set of perceptual vocabulary is called a Kansei semantic space. Additionally, D2DD properties are collected from the websites of several D2DD companies. Subsequently, a questionnaire is designed to survey customers' usage intentions and perceptual responses for D2DD services using the arranged perceptual vocabulary and D2DD properties. Finally, the decision tree is utilized to determine the degree of importance of each perceptual vocabulary word in terms of usage intention, and then generates several easily-explained rules which describe the relationship among usage intention, perceptual vocabulary, and D2DD properties. Such rules can be applied in D2DD design and improvement. Several objectives are hence shown as follows:

(1) Research and development departments can design a D2DD service based on the research findings. In addition, this study can be utilized to predict customers' usage intentions toward the D2DD before introducing it to the market, so that development risk and waste of resources are reduced.
(2) Marketing departments can propose some promotion strategies to enhance the perceived value of using D2DD according to the relationship between usage intention and perceptual responses.
(3) D2DD service providers may improve the existing services or develop new services based on the recommended service properties.

The remainder of this paper is organized as follows. The literature review about D2DD, Kansei Engineering, and decision tree is detailed in Section 2. The research framework, collection of D2DD properties and perceptual vocabulary, questionnaire design, sampling and data collection, and analytical tools are introduced in Section 3. Data analysis and analytical results are given in Section 4. Section 5 describes the conclusions with managerial implications. Research limitations and future research are discussed in Section 6.

2. Literature review

The following subsections introduce Kansei Engineering, decision tree, and service design on D2DD

2.1. Kansei Engineering and its procedure

Traditionally, product development depends on designers' sensibilities. However, these products may not be accepted by the market. Accordingly, it is important to get a better insight into customers' subjective perceptions. Nagamachi (1986) was the first to propose Kansei Engineering, which is a systematic product development approach to translate users' 'vibrations' into real design properties to satisfy their requirements. Kansei is a Japanese term which indicates human emotion, perceptual response, 'vibes/vibration', or impression, and its English meaning could be "user's psychological experience" in the domain of product development (Chuan, Sivaji, Shahimin, & Saad, 2013; Nagamachi, 1997; Schütte et al., 2004). Obviously, Kansei Engineering is a customer-oriented product development approach and can assist product designers in developing products fitting customers' requirements. Several Kansei Engineering expert systems integrating Genetic Algorithm (Tsuchiya, Matsubara, & Nagamachi, 1995) or Neural Network (Ishihara, Ishihara, Nagamachi, & Matsubara, 1995, 1997) are developed to simulate human image for the evaluation of product design. Matsubara and Nagamachi (1997) developed a Kansei Engineering system based on Linear Regression for door design. Nagamachi, Okazaki, and Ishikawa (2006) considered that perceptual responses do not always have linear features, and thus they integrated Kansei Engineering and rough sets theory for shoes design.

Since various types of Kansei Engineering represent different procedures, Schütte et al. (2004) proposed a conceptual framework for Kansei Engineering. The framework firstly determines the research domain and then describes the space of product properties and the Kansei semantic space. Subsequently, both spaces are merged with each other in the synthesis phase to show which product properties result in a perceptual response. The validation test can reveal the rationality of gathering the perceptual vocabulary in terms of the addressed product properties. Once the validated results are satisfactory, a recommendatory model is generated to indicate the relationship between the perceptual responses and product properties.

2.2. Decision tree

A decision tree is a data classification method. Compared with other classification methods, such as Logistic Regression, Artificial Neural Network, and Bayesian Classification, its output is represented as the form of "if… then", and thus it is easy to interpret and understand its analytical results (Murthy, 1998; Nie, Rowe, Zhang, Tian, & Shi, 2011). Moreover, Tan, Steinbach, and Kumar (2005) generalized several advantages of decision trees as follows:

(1) Decision tree induction is a nonparametric approach for building classification models.
(2) Constructing decision trees are inexpensive when the training set is large.
(3) Decision tree are easy to interpret.
(4) Decision tree algorithms are quite robust to the presence of noise, especially when methods for avoiding overfitting.
(5) The presence of redundant attributes does not adversely affect the accuracy of decision trees.

There are several famous decision tree algorithms such as ID3 (Quinlan, 1986), CART (Breiman, Olshen, & Stone, 1984), and C5.0 (Quinlan, 1996) in the literature. ID3 adopts the information gain measure, and thus prefers to select variables having a large number of values (Han, Kamber, & Pei, 2011). CART is efficient in terms of computational time, but its decision tree structure is very unstable (Özsoy & Şahin, 2009). C4.5 is a previous version of C5.0, and is an improved version from ID3. Compared with ID3 and C4.5, C5.0 adopts a boosting technology giving it better computational efficiency than C4.5, and can deal with class and continuous variables well (Bastı, Kuzey, & Delen, 2015; Pang & Gong, 2009). C5.0 also avoids the difficulties including over fitting for data and missing applicable data, and accordingly possesses a smaller tree and less memory consumption (Chiang, 2012). Several studies (Ali & Smith, 2006; Wong & Chung, 2007) have validated that C5.0 has better ability to identify crucial classification variables and to present appropriate decision-making rules.

Recently, the decision tree has been widely applied in many fields. Zuccaro (2010) used the decision tree on credit ratings and socio-demographic profiles to classify customers, and showed that it has great efficiency and explanatory power. Olson, Delen, and Meng (2012) studied prediction of bankruptcy and compared several data mining

tools. Their analytical results showed that C5.0 of IBM SPSS Modeler has better predictive accuracy than the other tools including ID3, C4.5 and CART. Chiang (2012) utilized the decision tree to establish online shoppers' markets and rules for dynamic CRM systems. Chuang, Chia, and Wong (2013) adopted the C5.0 decision tree to classify Taiwanese healthcare institutions based on customer value assessment, and developed marketing and relationship management strategies for each institution. Chiang, Tseng, and Torng (2013) employed the decision tree to analyze implant data to generate classification rules as a predictive model prior to implant surgery. Durvasula, Lysonski, and Mehta (2007) identified the set of attributes maximizing customers' satisfaction to the ocean freight shipping service by using a hybrid method, in which Neural Network is used to identify the relative importance of attributes and the decision tree is used to segment customers into several sub-groups based on their perceptions of various service features and their overall satisfaction. Chen, Hu, and Hsieh (2015) adopted the decision tree to construct a customer value model with length, recency, frequency, monetary, and profit factors in a logistics company to predict customer churn, and showed that the decision tree significantly outperforms Logistic Regression, Neural Network, and Support Vector Machine. Wu, Chen, and Tsau (2017) adopted the decision tree to develop cargo loss mitigation strategies for logistics management. They found that transit types, product categories, and shipping destinations are key factors behind cargo loss severity. Based on the great efficiency and explanatory power of the C5.0 decision tree, this study employs it to analyze the relationship between perceptual responses and product properties.

2.3. Importance of service design on D2DD

Service is a form of product that consists of activities, benefits, or satisfaction offered for sale which are essentially intangible and do not result in the ownership of anything (Kotler, 2003). D2DD is the last-mile home delivery and signifies that the goods are delivered to the designated address by a logistics service provider rather than customers picking up and shipping the goods in person (Chen et al., 2011). Therefore, D2DD is a kind of service according to Parasuraman, Zeithaml, and Berry (1988) and Chen et al. (2011). Moreover, D2DD service can be separated into three stages: package pick-up, package tracking, and package delivery (Chen, Hsu, et al., 2015).

Persson and Virum (2001) have stated that more and more users require fast, reliable, customized and cost-effective logistics processes and services. Several studies (Coyle, Bardi, & Langley, 2003; Liao et al., 2011; Mentzer & Williams, 2001) have mentioned that logistics service providers can gain a competitive advantage through customer-oriented service development, such as D2DD service development. Fadell et al. (2017) invented a smart home application that can recognize packages and sale drivers such that customers are unnecessary to stay at home for receiving. Danyluk et al. (2018) invented a package delivery tracking system to monitor the status of package delivery to avoid the package being stolen or tampered with. Both patents are developed based on the customer-oriented perspective. Allen, Thorne, and Browne (2007) emphasized that D2DD plays an important logistics component of order fulfillment, especially for e-commerce transactions, mail order, and purchases from TV shopping channels. Hsu, Lin, and Chen (2011) also mentioned that creating a personal and value-added D2DD service could promote the competitive advantage for logistics service providers. In addition, customers' requirements are always changing (Kano et al., 1984), that is, they always look forward to new services. Consequently, the logistics service providers should awaken to the importance of new service development to reflect customers' changes in preferences and needs.

Johne and Storey (1998) pointed out that new service design means innovation which directly or indirectly changes the form or quality of service. Gremyr, Witell, Löfberg, Edvardsson, and Fundin (2014) viewed innovation as changes in service characteristics. Salunke, Weerawardena, and McColl-Kennedy (2011) emphasized that service innovation can provide value for business partners. Yen, Wang, Wei, Hsu, and Chiu (2012) built and verified a structure of service innovation readiness and provided some insights into the management of service innovation by providing managers an assessment approach that can be used to gauge a firm's status, and direct its efforts in continuous improvement. Zeithaml, Parasuraman, and Berry (1990) argued that logistics service providers should constantly and actively explore new approaches to provide better services for their customers. Thus, for the purpose of service design, Dahlgaard, Schütte, Ayas, and Dahlgaard-Park (2008) suggested that Kansei Engineering could be utilized to construct the relationship between service properties and customers' perceptual responses and then generate a systematic procedure for service improvement and design. That is, D2DD service design and improvement could be achieved through Kansei Engineering.

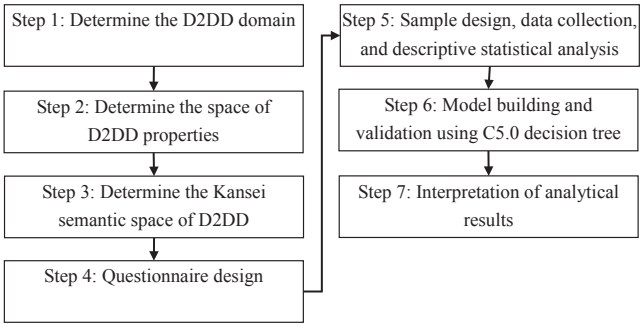


Fig. 1. The procedure of the proposed approach.

3. Methodology

This study develops a D2DD service design approach based on the framework proposed by Schütte et al. (2004) and data mining. Fig. 1 shows the procedure of the proposed approach, in which Steps 1–6 are illustrated in the following subsections and Step 7 are given in Section 4.

3.1. Step 1 - determine the D2DD domain

D2DD can be separated into two types based on temperature: ambient-temperature delivery and cold chain delivery. The former is a basic service for any D2DD service provider; the latter requires refrigerated devices to implement cold chain delivery. From the viewpoint of shippers, D2DD service can also be separated into B2C (business to customer) and C2C (customer to customer) deliveries. That is, shippers can be firms or individuals. Generally speaking, for B2C delivery, D2DD service providers are asked to provide customized services, and thus D2DD service providers should contract with shippers. C2C delivery is a relatively basic service provided by any D2DD service provider (Chen, Hsu, et al., 2015).

Due to the constant growth of the C2C e-commerce market, we select the C2C and ambient-temperature D2DD services as the research domain. In addition, D2DD users are categorized into shippers and receivers. The shippers are more active and contact the D2DD service process more than the receivers. Therefore, the shippers who have the usage experience of C2C and ambient-temperature D2DD in Taiwan are chosen as the respondents of Kansei survey.

3.2. Step 2 - determine the space of D2DD properties

Delivery of goods from a shipper to a receiver usually involves the following operations: collection, outbound, switching, inbound sorting, and delivery (Jung, Lee, & Chun, 2006). From the perspective of a service encounter, a D2DD company provides a package pick-up service

Table 1
Properties and property states of D2DD.

Stage of pick-up service
- E_1: Way to use service
  - P_1: *Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone, computer or app, and additionally can select the time of receiving package*
  - P_2: *Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone, computer or app, but cannot select the time of receiving package*
  - P_3: Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone and computer, and additionally can select the time of receiving package
  - P_4: Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone and computer, but cannot select the time of receiving package
  - P_5: Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone, and additionally can select the time of receiving package
  - P_6: Shipper sends package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone, but cannot select the time of receiving package
  - P_7: Shipper goes to the package receiving locations of D2DD service providers in person
- E_2: Cooperative package receiving locations
  - P_8: Convenience stores or other locations
  - P_9: Only convenience stores
  - P_{10}: No cooperative package receiving location
- E_3: Provides year-round service
  - P_{11}: Yes
  - P_{12}: No
- E_4: Provides packaging service
  - P_{13}: *Yes*
  - P_{14}: No
- E_5: Provides shipment protection plus service
  - P_{15}: Yes
  - P_{16}: No
- E_6: Provides protection of privacy
  - P_{17}: Yes
  - P_{18}: No
- E_7: Provides electronic consignment note
  - P_{19}: *Yes*
  - P_{20}: No
- E_8: Shipping fee calculation
  - P_{21}: *Single shipping fee calculation*
  - P_{22}: Multiple shipping fee calculation
- E_9: Provides reusable package to reduce shipping fees
  - P_{23}: Yes
  - P_{24}: No
- E_{10}: Payment mode
  - P_{25}: *Includes cash, electronic fare cards, and credit cards*
  - P_{26}: *Includes cash and electronic fare cards*
  - P_{27}: Cash only
- E_{11}: Payer
  - P_{28}: Shipper or receiver pays the shipping fee
  - P_{29}: Shipper pays the shipping fee

Stage of tracking service
- E_{12}: Package tracking service
  - P_{30}: *Shipper can track packages through computer, telephone, and app*
  - P_{31}: Shipper can undertake package tracking through telephone and computer
  - P_{32}: No package tracking service

Stage of delivery service
- E_{13}: Designate the delivery location
  - P_{33}: Shipper can designate the delivery address, the D2DD service provider's business locations, or its cooperative receiving locations
  - P_{34}: Shipper can designate the delivery address or the D2DD service provider's business locations
- E_{14}: Designate the arrival time
  - P_{35}: Shipper can designate a time period of a day and the specific date besides the same and the next dates
  - P_{36}: Shipper can designate a time period of a day
  - P_{37}: Shipper cannot designate a time period of a day and a date
- E_{15}: Promise to complete next-day delivery
  - P_{38}: Yes
  - P_{39}: No
- E_{16}: Provides same-day delivery service
  - P_{40}: *D2DD service provider actively provides same-day delivery service*
  - P_{41}: Shipper can ask the D2DD service provider to provide same-day delivery service
  - P_{42}: Without same-day delivery service
- E_{17}: Provides delivery service in the evening
  - P_{43}: Yes
  - P_{44}: No
- E_{18}: Cash on delivery (payment collection)
  - P_{45}: Receiver can pay to the D2DD service provider upon receipt of the package or pay by using the billing service provided by the D2DD service provider
  - P_{46}: Receiver can pay to the D2DD service provider upon receipt of the package
  - P_{47}: Cash on delivery is not allowed
- E_{19}: Receiver will be informed before starting the delivery service
  - P_{48}: Yes
  - P_{49}: No
- E_{20}: Receiver will be contacted when the package is going to arrive
  - P_{50}: Yes
  - P_{51}: No
- E_{21}: Shipper will be notified when the package arrives
  - P_{52}: Yes
  - P_{53}: No
- E_{22}: Continue to re-deliver the package a certain number of times
  - P_{54}: Yes
  - P_{55}: No
- E_{23}: D2DD service provider will actively contact the shipper if it encounter a problem, e.g. failing to deliver the package
  - P_{56}: Yes
  - P_{57}: No
- E_{24}: Allow the shipper to change the delivery address during the delivery stage
  - P_{58}: Yes
  - P_{59}: No
- E_{25}: Provides compensation measures
  - P_{60}: Yes
  - P_{61}: No
- E_{26}: Provides the complaint channel
  - P_{62}: Yes
  - P_{63}: No
- E_{27}: Environmental protection
  - P_{64}: Servicer provider considers environmental protection, energy saving, and carbon reduction
  - P_{65}: Servicer provider does not consider environmental protection, energy saving, and carbon reduction

to a customer when it implements the collection operation; the company provides a package tracking service to the customer when it implements outbound, switching, and inbound sorting operations; and the company provides a package delivery service to the customer when it implements delivery operations. Therefore, the D2DD service process is distinguished into three stages including package pick-up service, package tracking service, and package delivery service.

Subsequently, we collect D2DD properties and their relevant property states from the official websites and shipping forms of several famous D2DD service providers in Taiwan through discussion with several senior managers and experts. As the results show in Table 1, there are 27 properties and 65 corresponding property states including 9 creative property states marked in *italics*. The 27 service properties are divided into three categories of package pick-up service, package tracking service, and package delivery service. The notation E_i denotes the ith service property for $i = 1, 2, \ldots, 27$, and the notation P_j denotes the jth property state for $j = 1, 2, \ldots, 65$. Each property should display one of several property states, and each property state is a variable taking the value of either 0 (not offering the property state) or 1 (offering the property state). For instance, the service property of payment mode (i.e. E_{10}) can be one of the following states, P_{25}: including cash, electronic fare card, and credit card, P_{26}: including cash and electronic fare card, and P_{27}: cash only. If $P_{27} = 1$, then the D2DD service provider can only provide the cash payment, and P_{25} and P_{26} must be zero.

According to Nagamachi, Tachikawa, Imanishi, Ishizawa, and Yano (2008), it is suitable to select 10–20 stimuli (i.e. services) including a

Table 2
12 stimuli.

Property state	Stimulus (D2DD services)											
	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_{10}	S_{11}	S_{12}
P_1	1	0	0	0	0	0	0	0	0	0	1	0
P_2	0	0	0	0	0	0	0	0	1	0	0	1
P_3	0	1	0	0	0	0	0	0	0	0	0	0
P_4	0	0	0	0	0	0	1	0	0	0	0	0
P_5	0	0	0	0	0	1	0	1	0	0	0	0
P_6	0	0	1	0	1	0	0	0	0	0	0	0
P_7	0	0	0	1	0	0	0	0	0	1	0	0
P_8	1	0	0	1	1	0	0	0	0	0	0	1
P_9	0	1	0	0	0	1	1	0	0	0	1	0
P_{10}	0	0	1	0	0	0	0	1	1	1	0	0
P_{11}	1	1	1	0	0	0	1	1	0	0	1	0
P_{12}	0	0	0	1	1	1	0	0	1	1	0	1
P_{13}	1	1	1	1	1	1	0	0	0	0	0	0
P_{14}	0	0	0	0	0	0	1	1	1	1	1	1
P_{15}	1	0	0	0	0	0	0	1	1	1	1	1
P_{16}	0	1	1	1	1	1	1	0	0	0	0	0
P_{17}	1	1	0	0	0	0	1	1	1	1	0	0
P_{18}	0	0	1	1	1	1	0	0	0	0	1	1
P_{19}	1	1	1	0	1	1	1	1	0	1	1	0
P_{20}	0	0	0	1	0	0	0	0	1	0	0	1
P_{21}	1	0	0	0	1	1	0	0	0	1	0	0
P_{22}	0	1	1	1	0	0	1	1	1	0	1	1
P_{23}	1	1	0	0	0	1	1	1	0	0	0	1
P_{24}	0	0	1	1	1	0	0	0	1	1	1	0
P_{25}	1	0	1	1	0	0	0	0	0	0	0	1
P_{26}	0	0	0	0	1	0	0	0	1	1	1	0
P_{27}	0	1	0	0	0	1	1	1	0	0	0	0
P_{28}	1	1	1	0	0	1	1	1	0	1	1	0
P_{29}	0	0	0	1	1	0	0	0	1	0	0	1
P_{30}	1	1	0	0	0	1	1	0	0	0	0	1
P_{31}	0	0	1	0	0	0	0	1	1	1	0	0
P_{32}	0	0	0	1	1	0	0	0	0	0	1	0
P_{33}	1	1	1	0	0	1	1	1	0	1	1	0
P_{34}	0	0	0	1	1	0	0	0	1	0	0	1
P_{35}	1	0	0	0	1	0	0	0	0	1	1	0
P_{36}	0	0	0	0	0	1	1	1	0	0	0	1
P_{37}	0	1	1	1	0	0	0	0	1	0	0	0
P_{38}	1	0	1	0	1	1	0	1	1	0	0	1
P_{39}	0	1	0	1	0	0	1	0	0	1	1	0
P_{40}	1	0	0	0	1	1	0	0	0	1	0	0
P_{41}	0	1	0	1	0	0	1	0	0	0	1	0
P_{42}	0	0	1	0	0	0	0	1	1	0	0	1
P_{43}	1	0	1	0	1	0	1	0	1	0	1	0
P_{44}	0	1	0	1	0	1	0	1	0	1	0	1
P_{45}	1	0	0	1	0	0	1	0	0	1	0	0
P_{46}	0	1	0	0	1	0	0	1	0	0	1	0
P_{47}	0	0	1	0	0	1	0	0	1	0	0	1
P_{48}	1	1	1	0	0	0	0	1	1	1	1	1
P_{49}	0	0	0	1	1	1	1	0	0	0	0	0
P_{50}	1	0	0	1	1	1	0	1	0	1	1	0
P_{51}	0	1	1	0	0	0	1	0	1	0	0	1
P_{52}	1	0	1	0	1	0	1	1	1	0	1	0
P_{53}	0	1	0	1	0	1	0	0	0	1	0	1
P_{54}	1	0	1	0	0	1	0	1	0	1	1	0
P_{55}	0	1	0	1	1	0	1	0	1	0	0	1
P_{56}	1	0	0	1	1	1	0	1	0	1	0	1
P_{57}	0	1	1	0	0	0	1	0	1	0	1	0
P_{58}	1	0	0	1	0	0	1	1	1	1	0	1
P_{59}	0	1	1	0	1	1	0	0	0	0	1	0
P_{60}	1	0	1	0	1	0	1	0	1	0	1	0
P_{61}	0	1	0	1	0	1	0	1	0	1	0	1
P_{62}	1	0	1	0	1	0	1	0	1	0	1	0
P_{63}	0	1	0	1	0	1	0	1	0	1	0	1
P_{64}	1	1	0	1	0	1	0	1	0	1	0	1
P_{65}	0	0	1	0	1	0	1	0	1	0	1	0

creative stimulus for Kansei measurement. Therefore, for questionnaire design, these 27 properties, incorporating various property states, generate 12 stimuli denoted by S_1, S_2, …, S_{12} (refer to Table 2). For instance, the properties of stimulus S_1 is detailed in Table 3 by referring Tables 1 and 2. In the 12 stimuli, stimulus S_1, may be the highest quality D2DD service.

3.3. Step 3 - determine the Kansei semantic space of D2DD

The subsection describes the collection of perceptual vocabulary

Table 3
Property of stimulus S_1.

Property state	Description
P_1	Shipper can send package to the receiving locations of D2DD service providers in person, or asks the D2DD service providers to pick up package at the designated site by using telephone, computer or app, and additionally can select the time of receiving package
P_8	Its cooperative package receiving locations includes convenience stores or other locations
P_{11}	It provides year-round service
P_{13}	It provides packaging service
P_{15}	It provide shipment protection plus service
P_{17}	It provides protection of privacy
P_{19}	It provides electronic consignment note
P_{21}	Single shipping fee calculation
P_{23}	It provides reusable package to reduce shipping fees
P_{25}	Its payment modes includes cash, electronic fare cards, and credit cards
P_{28}	Shipper or receiver can pay the shipping fee
P_{30}	Shipper can track packages through computer, telephone, and app
P_{33}	Shipper can designate the delivery address, the D2DD service provider's business locations, or its cooperative receiving locations
P_{35}	Shipper can designate a time period of a day and the specific date besides the same and the next dates
P_{38}	It promises to complete next-day delivery
P_{40}	It actively provides same-day delivery service actively provides same-day delivery service
P_{43}	It provides delivery service in the evening
P_{45}	Receiver can pay to the D2DD service provider upon receipt of the package or pay by using the billing service provided by the D2DD service provider
P_{48}	Receiver will be informed before starting the delivery service
P_{50}	Receiver will be contacted when the package is going to arrive
P_{52}	Shipper will be notified when the package arrives
P_{54}	It continues to re-deliver the package a certain number of times
P_{56}	It will actively contact the shipper if it encounter a problem, e.g. failing to deliver the package
P_{58}	It allows the shipper to change the delivery address during the delivery stage
P_{60}	It provides compensation measures
P_{62}	It provides the complaint channel
P_{64}	It considers environmental protection, energy saving, and carbon reduction

Table 4
Original perceptual vocabulary.

Original perceptual vocabulary	High quality, on time, rapid, convenient, familiar, innovative, specialized, customized, kind, environmentally friendly, reliable, economical, time saving, immediate, expensive, cheap, complete, direct, general, true, excellent, trustworthy, uniform, omnipresent, flexible, new, transparent, elaborate, localized, efficient, safe, relieved, action-oriented

Note: Bold words are the perceptual vocabulary words which are preserved.

Table 5
38 perceptual vocabulary describing the three service stages.

Package pick-up service		Package tracking service		Package delivery service	
K_1	Rapid	K_{15}	Rapid	K_{26}	Rapid
K_2	Familiar	K_{16}	Familiar	K_{27}	Familiar
K_3	Specialized	K_{17}	Specialized	K_{28}	Specialized
K_4	On time	K_{18}	High quality	K_{29}	On time
K_5	High quality	K_{19}	Convenient	K_{30}	High quality
K_6	Convenient	K_{20}	Kind	K_{31}	Convenient
K_7	Kind	K_{21}	Immediate	K_{32}	Kind
K_8	Environmentally friendly	K_{22}	Reliable	K_{33}	Environmentally friendly
K_9	Immediate	K_{23}	Innovative	K_{34}	Immediate
K_{10}	Reliable	K_{24}	Time saving	K_{35}	Reliable
K_{11}	Innovative	K_{25}	Customized	K_{36}	Innovative
K_{12}	Economical			K_{37}	Time saving
K_{13}	Time saving			K_{38}	Customized
K_{14}	Customized				

(i.e. Kansei vocabulary). Based on the way of Llinares and Page (2011), we collect perceptual vocabulary (see Table 4) from different sources associated with D2DD, such as magazines, experts, newspapers, social networking sites, and advertisements on television, and then delete those redundant terms with similar meaning or which are determined through discussion with senior managers and experts to be less important for evaluation. Consequently, 14 perceptual vocabulary items including rapid, familiar, specialized, on time, high quality, convenient, kind, environmentally friendly, immediate, reliable, innovative, economical, time saving, and customized are retained. Furthermore, we expanded these 14 perceptual vocabulary words to 38 ones to tally with the package pick-up, package tracking, and package delivery services (refer to Table 5). For convenience, let K_t be the tth perceptual vocabulary word for $t = 1, 2, \ldots, 38$. The perceptual vocabulary words describing package pick-up (resp. tracking and delivery) service are K_1, $K_2, \ldots, K_{14}$ (resp. $K_{15}, K_{16}, \ldots, K_{25}$ and $K_{26}, K_{27}, \ldots, K_{38}$).

3.4. Step 4 - questionnaire design

The questionnaire is designed by referring to both D2DD property and Kansei semantic spaces, and is composed of the following parts.

Part 1 − demographic data: This part investigates the respondents' profiles. Several studies (Hsu, Chen, Chang, & Chao, 2010; Chen et al., 2011; Chen, Hsu, Hsu, & Lee, 2014) suggested that gender, age, occupation, education level, and income are appropriate demographic variables.

Part 2 − usage behavior of D2DD: The part investigates the usage behavior of D2DD. Each respondent is asked to indicate the D2DD service provider which he/she usually adopts, and his/her frequency of using the service.

Parts 3 and 4 − the first and second Kansei measurements of D2DD: We randomly choose 2 stimuli from the 12, bringing about a total of 66 combinations. Each questionnaire presents one combination of two different stimuli, where both stimuli are described respectively in Parts 3 and 4 to test the respondent's perceptions. In other words, there are in total 66 versions of the questionnaire. Each respondent is asked to give feedback on the 38 perceptual vocabulary and usage intention after reading one of the 66 versions. The usage intention and perceptual vocabulary are answered on a 3-point Likert scale (3 = agree, 2 = neutral, and 1 = disagree).

Since each respondent expresses the perceptual responses to two different stimuli, we can collect two instances of Kansei measurements from each respondent.

3.5. Step 5 - sample design, data collection, and descriptive statistical analysis

The primary respondents are the shippers who have the usage experience of C2C and ambient-temperature D2DD. For each version of the 66 questionnaires, one to three respondents are recruited. This study adopts an online survey to overcome time and space limits. The online questionnaire is produced by using the mySurvey system (http://www.mysurvey.tw/). Subsequently, a link to the online questionnaire is sent to the respondents via online forums of social networking sites (e.g., Facebook and Twitter), and e-mail. Each respondent' anonymity is protected. In order to promote the collection of effective samples, each respondent is asked to answer whether he/she had the usage experience of C2C and ambient-temperature D2DD before filling in Parts 1–4.

We receive a total of 162 completed questionnaires. According to the pretest results from 8 experienced users, one questionnaire should take approximately five to ten minutes for a respondent to complete. Thus, we eliminate questionnaires which are finished in less than five minutes. The number of retained questionnaires is 149. The effective response rate is thus 92%. Because each questionnaire includes two stimuli, the number of instances of Kansei measurements is 298. Descriptive statistical analysis is then adopted to present demographic data, information of D2DD usage behavior, and users' feelings.

3.6. Step 6 - model building and validation using C5.0 decision tree

The collected data are divided into two partitions: a training data set utilized to create a model and a testing data set to test the performance of the model, in which we utilize a decision tree to construct the model which describes the relationship among usage intention, perceptual vocabulary, and service properties.

In this study, a C5.0 decision tree of IBM SPSS Modeler 16.0 is used to construct the above mentioned relationship. C5.0 is utilized to predict which combinations of perceptual responses are crucial for usage intention and which combinations of service property states affect these crucial perceptual responses. As presented in Section 3.4, there are three classes (i.e. 3 = agree, 2 = neutral, and 1 = disagree) when treating the usage intention as the target variable. When treating one of the crucial perceptual vocabulary words as a target variable, its corresponding classes stand on the decision tree of usage intention.

Table 6
The statistical results of demographic data.

	Frequency	Percentage (%)
Gender		
Male	56	37.6%
Female	93	62.4%
Age		
Below 20	10	6.7%
20–30	94	63.1%
31–40	34	22.8%
41–50	9	6.0%
51 and over	2	1.3%
Education Level		
High school or below	8	5.4%
College	92	61.7%
Graduate school or above	49	32.9%
Occupation		
Public servants	12	8.1%
Manufacturing	20	13.4%
Business	36	24.2%
Unemployed (e.g. student, retired, housewife)	81	54.4%
Monthly income (NT$)		
Under 20,000	72	48.3%
20,001–30,000	13	8.7%
30,001–40,000	21	14.1%
40,001–50,000	22	14.8%
50,001–60,000	7	4.7%
Over 60,001	14	9.4%

Table 7
The statistical results of the D2DD usage behavior.

How long have you used D2DD service?	Frequency	Percentage (%)
Below 1 year	23	15.4%
1–2 years	37	24.8%
3–4 years	43	28.9%
5 years or over	46	30.9%
How frequently have you used D2DD service in the past year?		
Once or less	27	18.1%
2–3 times	71	47.7%
4–6 times	22	14.8%
7 times or more	29	19.5%
What kind of merchandise do you usually send?		
Book/documents	34	22.8%
Daily commodities	72	48.3%
Gifts	28	18.8%
Other	15	10.1%

C5.0 adopts an information gain ratio to evaluate splits at each internal node of a tree (Mistikoglu et al., 2015). A variable with the highest information gain ratio is selected as a split node. Considering a training set D partitioned from the data obtained via the questionnaire survey, the information gain ratio of variable V (e.g. perceptual vocabulary word or property state), denoted by $GainRatio(V)$, is calculated by the following equation:

$$GainRatio(V) = Gain(V)/Split_V(D), \tag{1}$$

where $Gain(V)$ represents the information gained by splitting D using V, and $Split_V(D)$ represents a split information value. In order to obtain $GainRatio(V)$, we should calculate $Gain(V)$ and $Split_V(D)$. Let w be the number of classes, $|D|$ be the number of instances in the training set D, and $freq(G_u, D)$ be the number of the instances belonging to the ith class in D, where G_u is the uth class for $u = 1, 2, \ldots, w$. Thus, $p_i = freq(G_u, D)/|D|$ represents the probability of instances in D belonging to the uth class. The expected information to classify a tuple in D is given by

$$Inf\ (D) = -\sum_{u=1}^{w} p_u \log_2 p_u. \tag{2}$$

According to the variable V with m values, we can divide D into m partitions including D_1, D_2, …, and D_m. The entropy after applying the variable V, denoted by $inf_V(D)$, is calculated via the following equation:

$$Inf_V(D) = -\sum_{r=1}^{m} \frac{|D_r|}{|D|} \times inf\,(D_r), \tag{3}$$

where $\frac{|D_r|}{|D|}$ means the weight of the rth partition. Therefore, both $Gain(V)$ and $Split_V(D)$ are represented as the following equations:

$$Gain(V) = inf\,(D) - inf_V(D); \tag{4}$$

$$Split_V(D) = -\sum_{r=1}^{m} \frac{|D_r|}{|D|} \times inf\,(D_r). \tag{5}$$

4. Results and analysis of D2DD case

This section describes and validates the analytical results of collected data from descriptive statistics and C5.0, and then discusses the obtained results.

Table 8
The average scores and standard deviations of perceptual responses and usage intention.

	Stimulus											
	S_1		S_2		S_3		S_4		S_5		S_6	
Variable	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD
K_1	**2.65**	0.573	2.550	0.686	1.926	0.675	1.79	0.721	2.00	0.780	2.44	0.751
K_2	**2.83**	0.388	2.600	0.681	2.000	0.734	1.63	0.770	2.04	0.806	2.41	0.747
K_3	**2.91**	0.288	2.650	0.671	2.000	0.784	1.75	0.676	2.08	0.717	2.67	0.620
K_4	**2.78**	0.422	2.550	0.605	2.148	0.602	1.71	0.624	2.13	0.741	2.44	0.641
K_5	**3.00**	0.000	2.550	0.686	1.889	0.698	1.75	0.608	1.88	0.741	2.59	0.572
K_6	**2.96**	0.209	2.600	0.598	1.815	0.879	1.96	0.859	2.00	0.834	2.59	0.747
K_7	**2.83**	0.491	2.350	0.745	1.741	0.712	1.75	0.737	1.75	0.676	2.33	0.679
K_8	**2.78**	0.422	2.550	0.605	1.815	0.786	1.54	0.721	1.88	0.680	2.33	0.734
K_9	**2.83**	0.388	2.550	0.686	1.889	0.801	1.63	0.711	2.13	0.680	2.44	0.801
K_{10}	**2.83**	0.491	2.700	0.571	1.593	0.797	1.67	0.761	1.96	0.690	2.30	0.724
K_{11}	**2.74**	0.619	2.200	0.696	1.556	0.698	1.50	0.722	1.83	0.702	2.07	0.781
K_{12}	**2.78**	0.518	2.100	0.788	1.704	0.669	1.63	0.576	1.92	0.717	2.41	0.694
K_{13}	**2.91**	0.288	2.450	0.686	1.926	0.781	1.58	0.717	2.00	0.659	2.37	0.792
K_{14}	**2.78**	0.518	2.500	0.761	1.815	0.786	1.42	0.654	1.63	0.647	2.48	0.753
K_{15}	**2.87**	0.458	2.600	0.598	2.407	0.747	1.42	0.776	1.42	0.654	2.56	0.698
K_{16}	**2.83**	0.491	2.700	0.470	2.778	0.424	1.29	0.624	1.17	0.482	2.74	0.656
K_{17}	**2.83**	0.491	2.700	0.571	2.704	0.542	1.38	0.647	1.29	0.550	2.81	0.483
K_{18}	**2.87**	0.458	2.700	0.470	2.704	0.542	1.38	0.711	1.25	0.532	2.70	0.669
K_{19}	2.87	0.458	2.800	0.410	2.741	0.656	1.42	0.717	1.21	0.509	2.78	0.577
K_{20}	**2.70**	0.559	2.400	0.598	2.519	0.643	1.33	0.637	1.21	0.509	2.59	0.694
K_{21}	**2.91**	0.288	2.850	0.366	2.667	0.480	1.33	0.637	1.21	0.509	2.59	0.747
K_{22}	**2.87**	0.344	2.700	0.571	2.667	0.555	1.38	0.647	1.21	0.509	2.48	0.802
K_{23}	2.61	0.722	**2.650**	0.489	2.000	0.734	1.42	0.717	1.29	0.624	2.44	0.751
K_{24}	**2.87**	0.458	2.750	0.550	2.481	0.509	1.46	0.779	1.25	0.532	2.63	0.688
K_{25}	**2.74**	0.541	2.400	0.681	2.444	0.698	1.42	0.717	1.17	0.482	2.48	0.802
K_{26}	**2.83**	0.491	2.050	0.759	2.000	0.679	1.58	0.584	2.13	0.797	2.56	0.698
K_{27}	**3.00**	0.000	1.650	0.671	1.741	0.813	1.38	0.495	1.88	0.741	2.22	0.847
K_{28}	**3.00**	0.000	2.100	0.718	1.852	0.770	1.63	0.647	2.13	0.797	2.44	0.698
K_{29}	**2.91**	0.417	2.150	0.745	1.963	0.706	1.58	0.717	2.17	0.816	2.41	0.747
K_{30}	**3.00**	0.000	1.700	0.733	1.593	0.694	1.46	0.509	2.00	0.780	2.26	0.859
K_{31}	**2.96**	0.209	1.700	0.733	1.778	0.751	1.42	0.654	2.00	0.885	2.26	0.813
K_{32}	**2.87**	0.344	1.850	0.745	1.593	0.797	1.42	0.584	1.79	0.779	2.33	0.734
K_{33}	**2.87**	0.344	2.200	0.768	1.370	0.565	2.08	0.830	1.58	0.776	2.48	0.753
K_{34}	**2.87**	0.458	1.900	0.718	1.889	0.751	1.46	0.588	2.08	0.830	2.37	0.839
K_{35}	**2.91**	0.417	1.750	0.716	1.889	0.801	1.33	0.482	2.13	0.797	2.33	0.734
K_{36}	**2.74**	0.541	1.800	0.616	1.444	0.641	1.42	0.504	1.92	0.830	1.96	0.649
K_{37}	**2.87**	0.344	1.850	0.745	1.556	0.698	1.46	0.658	1.96	0.859	2.30	0.823
K_{38}	**2.78**	0.518	1.750	0.716	1.630	0.688	1.42	0.584	1.79	0.779	2.11	0.801
UI	**2.96**	0.209	1.950	0.759	1.481	0.580	1.46	0.588	1.83	0.761	2.15	0.818
Perceptive Avg.	**2.85**	0.430	2.33	0.739	2.01	0.795	1.53	0.682	1.75	0.774	2.44	0.741

	Stimulus											
	S_7		S_8		S_9		S_{10}		S_{11}		S_{12}	
Variable	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD
K_1	2.29	0.806	2.58	0.654	2.29	0.690	1.96	0.706	2.63	0.565	2.22	0.751
K_2	2.46	0.658	2.58	0.584	1.88	0.900	1.70	0.823	2.48	0.700	2.19	0.786
K_3	2.71	0.550	2.50	0.722	2.25	0.897	2.26	0.764	2.63	0.565	2.30	0.823
K_4	2.42	0.776	2.46	0.658	2.33	0.637	2.33	0.734	2.56	0.577	2.22	0.847
K_5	2.46	0.658	2.67	0.565	2.25	0.737	2.11	0.801	2.37	0.629	2.22	0.751
K_6	2.42	0.717	2.67	0.637	2.13	0.947	1.67	0.832	2.70	0.465	2.33	0.832
K_7	2.33	0.761	2.46	0.588	2.08	0.776	1.67	0.620	2.59	0.572	2.11	0.751
K_8	2.58	0.654	2.25	0.737	1.71	0.908	1.89	0.847	2.15	0.770	2.41	0.747
K_9	2.46	0.721	2.50	0.659	1.96	0.806	1.74	0.859	2.48	0.643	2.37	0.688
K_{10}	2.67	0.565	2.42	0.584	2.17	0.868	2.33	0.784	2.33	0.620	2.26	0.764
K_{11}	2.38	0.711	2.21	0.779	2.17	0.917	1.85	0.864	2.33	0.734	1.93	0.616
K_{12}	2.46	0.721	2.33	0.702	1.92	0.776	1.78	0.751	2.11	0.801	2.15	0.718
K_{13}	2.42	0.717	2.42	0.654	1.79	0.884	1.78	0.801	2.52	0.700	2.15	0.818
K_{14}	2.42	0.654	2.42	0.717	1.67	0.917	1.67	0.734	2.33	0.784	2.04	0.759
K_{15}	2.83	0.381	2.63	0.647	2.67	0.565	2.37	0.742	1.70	0.775	2.44	0.751
K_{16}	2.75	0.442	2.75	0.532	2.71	0.624	2.44	0.801	1.67	0.832	2.56	0.641
K_{17}	2.83	0.381	2.58	0.654	2.71	0.624	2.44	0.698	1.67	0.832	2.52	0.643
K_{18}	2.79	0.415	2.63	0.576	2.58	0.717	2.44	0.751	1.67	0.832	2.48	0.700
K_{19}	**2.88**	0.338	**2.88**	0.448	2.75	0.608	2.37	0.792	1.74	0.903	2.59	0.694
K_{20}	2.58	0.584	2.46	0.658	2.58	0.584	2.37	0.742	1.63	0.792	2.33	0.784
K_{21}	2.83	0.381	2.79	0.509	2.75	0.608	2.37	0.792	1.67	0.832	2.52	0.802
K_{22}	2.67	0.482	2.63	0.647	2.67	0.637	2.48	0.700	1.67	0.832	2.41	0.747

(*continued on next page*)

Table 8 (*continued*)

	Stimulus											
	S_1		S_2		S_3		S_4		S_5		S_6	
Variable	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD	Avg.	SD
K_{23}	2.54	0.588	2.42	0.654	2.50	0.659	2.04	0.808	1.74	0.859	2.15	0.718
K_{24}	2.58	0.584	2.58	0.584	2.67	0.482	2.33	0.679	1.89	0.892	2.41	0.694
K_{25}	2.46	0.721	2.58	0.584	2.33	0.761	2.26	0.859	1.81	0.879	2.19	0.786
K_{26}	2.54	0.721	2.54	0.658	2.38	0.770	2.56	0.641	2.30	0.823	2.19	0.879
K_{27}	2.38	0.770	2.50	0.659	2.29	0.859	2.44	0.801	1.96	0.898	2.11	0.892
K_{28}	2.42	0.654	2.58	0.654	2.38	0.770	2.56	0.751	2.15	0.864	2.19	0.786
K_{29}	2.50	0.659	2.58	0.584	2.25	0.794	2.59	0.636	2.26	0.764	2.11	0.698
K_{30}	2.38	0.711	2.54	0.588	2.21	0.884	2.48	0.700	2.00	0.832	2.15	0.770
K_{31}	2.33	0.816	2.46	0.721	2.33	0.816	2.33	0.784	2.07	0.781	2.07	0.829
K_{32}	2.29	0.806	2.13	0.797	2.29	0.806	2.37	0.792	1.93	0.781	2.00	0.877
K_{33}	2.08	0.830	2.42	0.654	1.54	0.721	2.37	0.742	1.74	0.712	2.37	0.792
K_{34}	2.33	0.761	2.42	0.654	2.21	0.833	2.44	0.751	2.11	0.801	2.22	0.801
K_{35}	2.38	0.824	2.29	0.751	2.33	0.868	2.56	0.641	2.00	0.832	2.30	0.869
K_{36}	2.08	0.717	2.17	0.761	1.88	0.797	2.19	0.786	1.67	0.620	1.96	0.759
K_{37}	2.42	0.776	2.33	0.702	2.17	0.868	2.37	0.742	2.04	0.759	2.04	0.854
K_{38}	2.17	0.868	2.42	0.654	1.96	0.908	2.33	0.784	1.85	0.770	2.00	0.832
UI	2.54	0.721	2.50	0.590	2.21	0.884	2.26	0.764	2.11	0.751	2.19	0.834
Perceptive Avg.	2.49	0.685	2.49	0.658	2.26	0.825	2.22	0.803	2.08	0.819	2.24	0.778

Note: Avg. represents the average score; SD represents the standard deviation; UI represents the usage intention; Perceptive Avg. is the average score of all perceptual responses for a stimulus. Bold means the average score is the highest.

Table 9
Comparison among various minimum records per child branch.

	Minimum instances per child branch		
	$\theta = 2$	$\theta = 7$	$\theta = 10$
Average tree depth	10.2	4	3.4
Average number of nodes	51.7	10	8.2
Average training accuracy	86.326%	72.018%	70.383%
Average testing accuracy	62.808%	64.394%	63.71%

Table 10
Comparison between the decision tree and the logistic regression.

	Decision tree	Logistic regression
Average training accuracy	72.018%	71.809%
Average testing accuracy	64.394%	63.675%

4.1. Sample profile

The profile of respondents is summarized in Table 6 (i.e. the descriptive statistics of Part 1 in the questionnaire). The sample profile is similar to the one of a Taiwanese D2DD user experience survey from Pollster Technology Marketing Ltd (see http://www.pollster.com.tw/Aboutlook/lookview_item.aspx?ms_sn=2632#). In terms of gender, there are 1999 respondents in their sample, in which 54.2% of respondents are female (the percentage in Table 6 is 62.4%). In terms of age, the percentage of age 20–50 years is 90.8% (the percentage in Table 6 is 91.9%). Thus, our sample is representative. Table 7 summarizes the statistical results of D2DD usage behavior. Most of the respondents have used D2DD for at least one year (84.6%). The percentage of the respondents using D2DD 2–3 times within one year is 47.6%. The commonly delivered merchandise is daily commodities (48.3%), followed by books and documents (22.8%).

Table 8 describes the average scores and standard deviations of perceptual vocabulary and usage intention for each stimulus. Obviously, the quality stimulus S_1 performs very well in terms of most perceptual vocabulary except for K_{19} and K_{23}. Stimuli S_4 and S_5 perform poorly in terms of usage intention and perceptive average score, since most of the perceptual responses are negative. Stimuli, $S_1 - S_3$, $S_6 - S_{12}$, have positive responses on most of the perceptual vocabulary, that is, their perceptive average scores exceed 2. However, both stimuli S_2 and S_3 have negative responses on usage intention. Apparently, there should exist some important perceptual vocabulary which has a great impact on usage intention.

4.2. The perceptual responses related to D2DD usage intention

The data are partitioned, with approximately 2/3 used for the training set and 1/3 for the testing set. Usage intention is regarded as a target variable, and the perceptual vocabulary words are input variables. Before analyzing the relationship between perceptual responses and usage intention, the performance of the decision tree is discussed from the viewpoint of accuracy. Firstly, the effect of minimum records per child branch of decision tree modeling on accuracy is explored, where minimum records per child branch is denoted by θ and is employed to limit the number of splits in any branch of the tree. Three settings including $\theta = 2$, $\theta = 7$, and $\theta = 10$ are taken into account. Each setting is executed for 10 experiments to observe the average depth, the average number of nodes, the average training accuracy, and the average testing accuracy, where the training and testing sets the three decision trees utilize in each experiment are different. The results are displayed in Table 9. Significantly, the decision tree with $\theta = 2$ is overfitting and complicated. The decision tree with $\theta = 10$ is easier to understand due to the least depth and nodes but its accuracy performs worse than the decision tree with $\theta = 7$.

In addition, the decision tree is compared with a logistic regression, which is powerful in classification problems (Chen, Hsu, et al., 2015; Nie et al., 2011). This study uses the logistic regression of IBM SPSS Modeler 16.0. Both methods are also required to be implemented 10 experiments to obtain the average training and testing accurate rates. The comparison result is shown in Table 10, where the experimental results the decision tree with $\theta = 7$ is adopted. The average training (resp. testing) accuracy of the decision tree is reasonable at 72.018% (resp. 64.394%), and is slightly higher than and the average training (resp. testing) accuracy 71.809% (resp. 63.675%) of the logistic regression. Moreover, the results from the decision tree is easier explained and understood since the logistic regression does not

In accordance with the above experiments, the decision tree with $\theta = 7$ has reasonable accuracy and is easily interpreted. In its 10

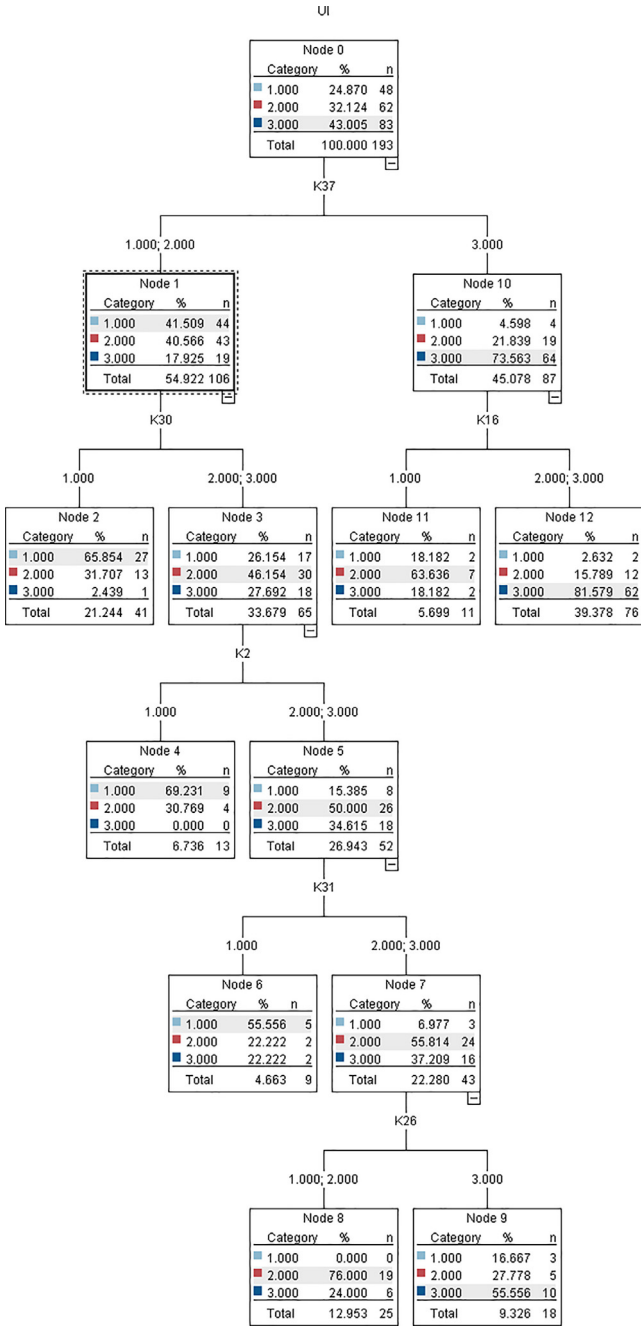


Fig. 2. Decision tree of usage intention.

experiments, the decision tree has the smallest difference between the training accuracy and the testing accuracy is applied and shown in Fig. 2. The generated rules are listed in Table 11, where the training (resp. testing) accuracy is reasonable at 72.02% (resp. 70.48%). According to Table 11, there are six important perceptual vocabulary words, including K_2, K_{16}, K_{26}, K_{30}, K_{31}, and K_{37}, related to the usage intention of D2DD service, and most of the users are concerned about how it feels to use a package delivery service. Subsequently, each important perceptual vocabulary word is also regarded as a target variable, and all of its corresponding property states are input variables, with each perceptual vocabulary word classified into two classes according to the responses in Table 11. Since the above comparison between the decision tree and the logistic regression are explored through the relationship analysis between perceptual responses and usage intention, this stage of analyzing the relationship between perceptual responses and service property states does not take such comparisons into account. Six decision trees of the important perceptual vocabulary words are presented in Fig. 3. Table 12 shows the corresponding rules describing the relationship between the D2DD property states and the important perceptual vocabulary. Each perceptual vocabulary word's decision tree model reasonably exceeds 57% accuracy and is not overfitting. Table 13 summarizes the importance of perceptual vocabulary words to usage intention as well as the importance of service property states to the perceptual vocabulary words, where the importance is evaluated by the SPSS C5.0 decision tree. The information provides some aspects of D2DD service design or improvement.

Table 11
Decision rules of usage intention.

Decision tree for UI (72.02% training accuracy; 70.48% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
1	If $K_{37} = = 3$ and $K_{16} \geq 2$, then UI = = 3	(76, 39.38%, 0.816)
2	If $K_{37} \leq 2$ and $K_{30} \geq 2$ and $K_2 \geq 2$ and $K_{31} \geq 2$ and $K_{26} = = 3$, then UI = = 3	(18, 9.33%, 0.556)
3	If $K_{37} \leq 2$ and $K_{30} = = 1$, then UI = = 1	(41, 21.24%, 0.659)
4	If $K_{37} \leq 2$ and $K_{30} \geq 2$ and $K_2 = = 1$, then UI = = 1	(13, 6.74%, 0.692)
5	If $K_{37} \leq 2$ and $K_{30} \geq 2$ and $K_2 \geq 2$ and $K_{31} = = 1$, then UI = = 1	(9, 4.66%, 0.556)
6	If $K_{37} \leq 2$ and $K_{30} \geq 2$ and $K_2 \geq 2$ and $K_{31} \geq 2$ and $K_{26} \leq 2$, then UI = = 2	(25, 12.95%, 0.760)
7	If $K_{37} = = 3$ and $K_{16} = = 1$, then UI = = 2	(11, 5.70%, 0.636)

Note: UI represents "usage intention"; K_2 represents "pick-up service is familiar."; K_{16} represents "tracking service is familiar."; K_{26} represents "delivery service is rapid"; K_{30} represents "delivery service is high quality"; K_{31} represents "delivery service is convenient"; K_{37} represents "delivery service is time saving"; 3 means "agree"; 2 means "neutral"; 1 means "disagree".

4.2.1. Analysis for the positive usage intention

Observe rules 1 and 2 with positive usage intention (i.e. UI = = 3) in Table 11. It is better to make $K_{37} = = 3$ and $K_{16} \geq 2$ because rule 1 has higher coverage of instances and greater accuracy than that of rule 2. Based on Table 12, the crucial D2DD property states should be $P_{37} = = 0$ and $P_{33} = = 1$ to achieve $K_{37} = = 3$. Similarly, to achieve $K_{16} \geq 2$, the crucial D2DD property state should be $P_{32} = = 0$. Accordingly, the only property combination which may result in rule 1 is recommended in Table 14, i.e. $P_{37} = = 0$, $P_{33} = = 1$ and $P_{32} = = 0$. Such a combination considered in D2DD design emphasizes several customized services including the location and time designation and various forms of package tracking. Although no creative property state is included in the property combination, $P_{32} = = 0$ implies that the D2DD service provider can provide a package tracking service not only through computer or by phone, but also a creative software application (app).

Rule 2 suggests that a D2DD service provider can make $K_{37} \leq 2$, $K_{30} \geq 2$, $K_2 \geq 2$, $K_{31} \geq 2$, and $K_{26} = = 3$ such that UI = = 3. Based on Table 12, we find that there is no feasible property combination to satisfy the perceptual responses in rule 2. For instance, there exists a possible property combination requiring $P_{37} = = 1$ to satisfy $K_{37} \leq 2$, $P_{37} = = 0$ to satisfy $K_{30} \geq 2$, $P_7 = = 0$ to satisfy $K_2 \geq 2$, $P_{37} = = 0$ to satisfy $K_{31} \geq 2$, and $P_{37} = = 0$ and $P_{33} = = 1$ to satisfy $K_{26} = = 3$, simultaneously. Obviously, such a property combination consists of conflict property states, i.e. $P_{37} = = 1$ and $P_{37} = = 0$. In other words, there is no recommendation of D2DD design for rule 2.

4.2.2. Analysis for the negative usage intention

Rules 3–5, in Table 11, describe the perceptual responses resulting in the negative usage intention (i.e. UI = = 1). Rule 3 suggests that any D2DD service provider should not make $K_{37} \leq 2$ and $K_{30} = = 1$

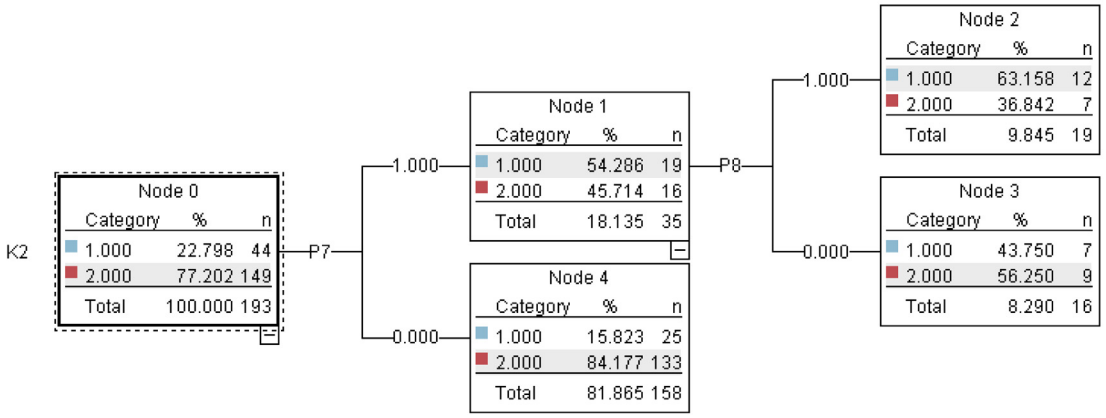
simultaneously, in which $K_{37} \leq 2$ results from $P_{37} = = 1$ and $K_{30} = = 1$ results from $P_{37} = = 1$ and $P_{45} = = 1$ (refer to Table 12). The corresponding property combination for rule 3 are shown in Table 15. The property state of $P_{37} = = 1$ is obviously worse than the other states of "E_{14}: designate the arrival time", but $P_{45} = = 1$ is the best property state from the viewpoint of "E_{18}: cash on delivery". That is, the combination $P_{37} = = 1$ and $P_{45} = = 1$ may result in $K_{37} \leq 2$ and $K_{30} = = 1$ such that UI = = 1, and is not recommended to be included in D2DD design.

Rule 4 reveals that the negative usage intention may result from $K_{37} \leq 2$, $K_{30} \geq 2$, and $K_2 = = 1$. Based on Tables 12 and 16, such a situation may be from property combinations 3 and 4 consisting of some bad property states (e.g. $P_7 = = 1$ and $P_{37} = = 1$ for combination 3 and $P_7 = = 1$ and $P_{33} = = 0$ for rule 4) and a good property state (e.g. $P_8 = = 1$). We hence conclude that both property combinations are not recommended in D2DD design.

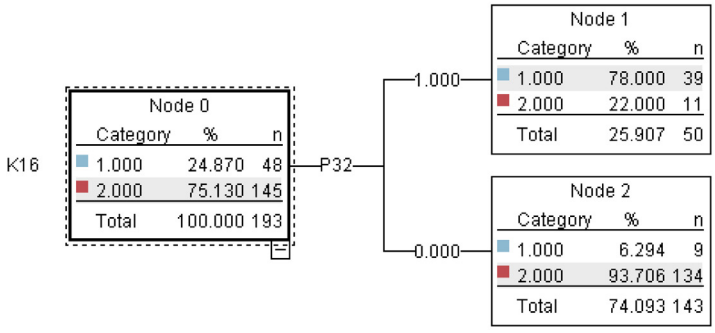
Rule 5 shows that the negative usage intention may result from $K_{37} \leq 2$, $K_{30} \geq 2$, $K_2 \geq 2$, and $K_{31} = = 1$. However, there is no feasible property combination due to the conflict property states. Therefore, there exists no non-recommended D2DD design for rule 5.

4.2.3. Analysis for the neutral usage intention

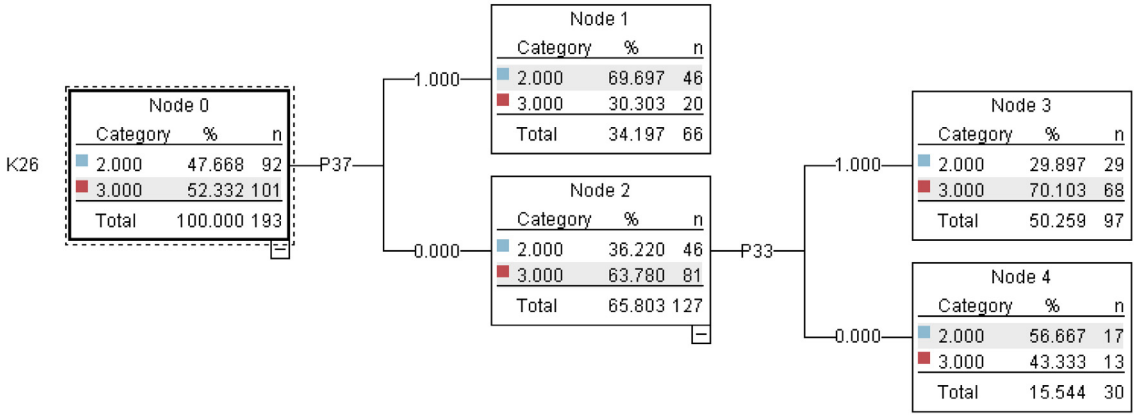
Rules 6 and 7 describe the neutral usage intention, and its related perceptual responses. The feasible property combinations for rule 6 (resp. 7) are given in Table 17 (resp. Table 18). Comparing rule 6 with rule 2, the respondents grouped into rule 6 may agree with usage



(a) Decision tree of K_2

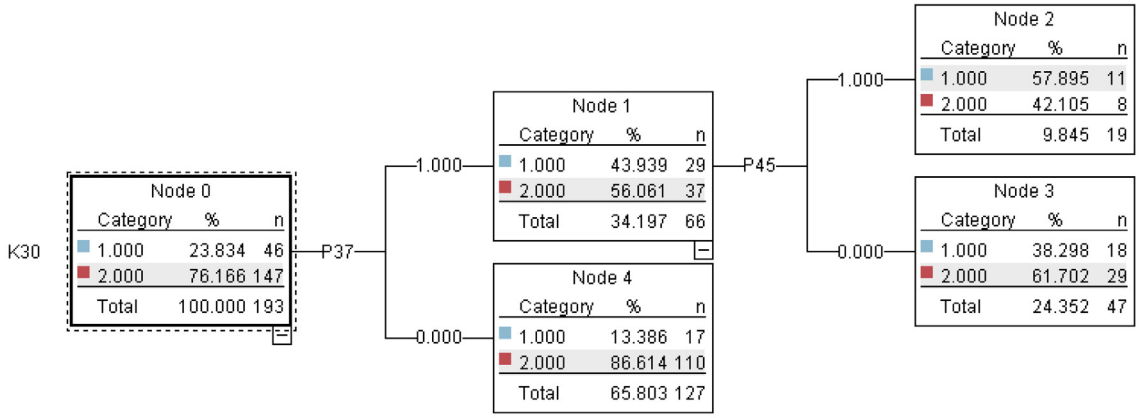


(b) Decision tree of K_{16}

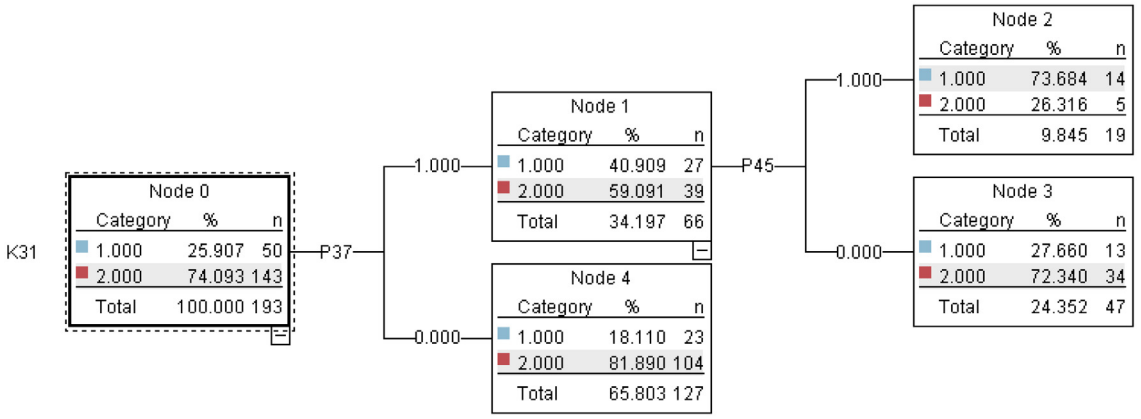


(c) Decision tree of K_{26}

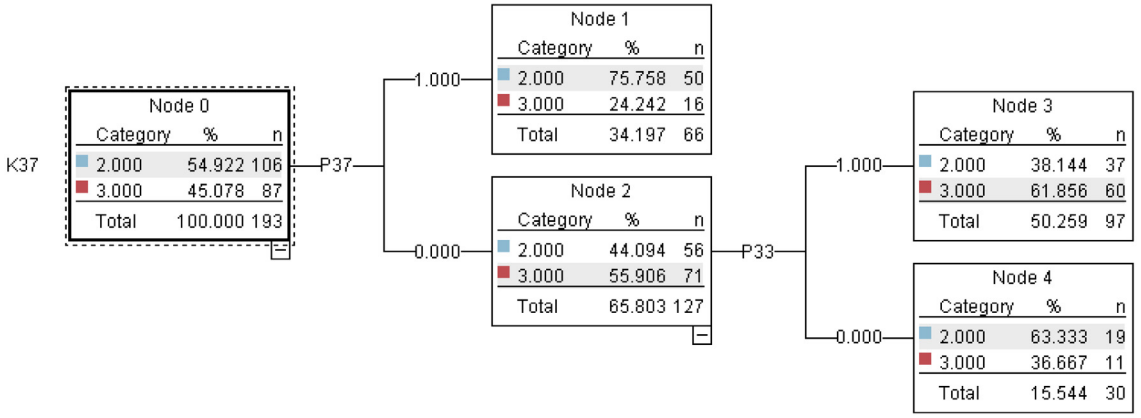
Fig. 3. Decision trees of important perceptual vocabulary.



(d) Decision tree of K_{30}



(e) Decision tree of K_{31}



(f) Decision tree of K_{37}

Fig. 3. (*continued*)

intention by improving their perceptual response from $K_{26} \leq 2$ to $K_{26} = = 3$ under $K_{37} \leq 2$, $K_{30} \geq 2$, $K_2 \geq 2$, and $K_{31} \geq 2$. Such an improvement seems to be achieved by changing $P_{37} = = 1$ (resp. $P_{37} = = 0$ and $P_{33} = = 0$) to $P_{37} = = 0$ and $P_{33} = = 1$ such that $K_{37} = = 3$ and $K_{26} = = 3$ for property combinations 6 and 7 (resp. property combinations 8 and 9). Similarly, the respondents grouped into rule 7 may have positive usage intention by improving their perceptual response from $K_{16} = = 1$ to $K_{16} \geq 2$ under $K_{37} = = 3$. In order to achieve this improvement, the D2DD service provider can provide the tracking service in D2DD design via telephone, computer, or mobile app, i.e. let $P_{32} = = 0$.

4.2.4. Analysis for the stimuli

By observing Tables 12 and 13, the crucial property states which are

Table 12
Decision rules of important perceptual vocabulary.

Rule No.	Rule description	(No. of instances,% of instances, accuracy)
Decision tree for K_{37} (66.84% training accuracy; 57.14% testing accuracy)		
(1)	If $P_{37} = = 1$, then $K_{37} \leq 2$	(66, 34.20%, 0.758)
(2)	If $P_{37} = = 0$ and $P_{33} = = 1$, then $K_{37} = = 3$	(97, 50.26%, 0.619)
(3)	If $P_{37} = = 0$ and $P_{33} = = 0$, then $K_{37} \leq 2$	(30, 15.54%, 0.633)
Decision tree for K_{30} (77.72% training accuracy; 67.62% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
(1)	If $P_{37} = = 1$ and $P_{45} = = 1$, then $K_{30} = = 1$	(19, 9.84%, 0.579)
(2)	If $P_{37} = = 1$ and $P_{45} = = 0$, then $K_{30} \geq 2$	(47, 24.35%, 0.617)
(3)	If $P_{37} = = 0$, then $K_{30} \geq 2$	(127, 65.80%, 0.866)
Decision tree for K_{16} (89.64% training accuracy; 86.67% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
(1)	If $P_{32} = = 1$, then $K_{16} = = 1$	(50, 25.91%, 0.780)
(2)	If $P_{32} = = 0$, then $K_{16} \geq 2$	(143, 74.09%, 0.937)
Decision tree for K_{31} (78.76% training accuracy; 66.67% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
(1)	If $P_{37} = = 1$ and $P_{45} = = 1$, then $K_{31} = = 1$	(19, 9.84%, 0.737)
(2)	If $P_{37} = = 1$ and $P_{45} = = 0$, then $K_{31} \geq 2$	(47, 24.35%, 0.723)
	If $P_{37} = = 0$, then $K_{31} \geq 2$	(127, 65.80%, 0.819)
Decision tree for K_2 (79.79% training accuracy; 72.38% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
(1)	If $P_7 = = 1$ and $P_8 = = 1$, then $K_2 = = 1$	(19, 9.84%, 0.632)
(2)	If $P_7 = = 1$ and $P_8 = = 0$, then $K_2 \geq 2$	(16, 8.29%, 0.562)
(3)	If $P_7 = = 0$, then $K_2 \geq 2$	(158, 81.87%, 0.842)
Decision tree for K_{26} (67.88% training accuracy; 63.81% testing accuracy)		
Rule No.	Rule description	(No. of instances,% of instances, accuracy)
(1)	If $P_{37} = = 1$, then $K_{26} \leq 2$	(66, 34.20%, 0.697)
(2)	If $P_{37} = = 0$ and $P_{33} = = 1$, then $K_{26} = = 3$	(97, 50.26%, 0.701)
(3)	If $P_{37} = = 0$ and $P_{33} = = 0$, then $K_{26} \leq 2$	(30, 15.54%, 0.567)

Note: P_7 represents "shipper can only go to the package receiving locations of D2DD service providers in person"; P_8 represents "cooperative package receiving locations include convenience stores or other locations"; P_{32} represents "no package tracking service"; P_{33} represents "shipper can designate the delivery address, the D2DD service provider's business locations, or its cooperative receiving locations"; P_{37} represents "shipper cannot designate a time period of a day and a date"; P_{45} represents "receiver can pay to the D2DD service provider upon receipt of the package or pay by using the billing service provided by the D2DD service provider"; '1' represents service with this property state; '0' represents service without this property state.

related to the crucial perceptual vocabulary are P_7, P_8, P_{32}, P_{33}, P_{37}, and P_{45}, and then the best property combination should consist of $P_7 = = 0$, $P_8 = = 0$, $P_{32} = = 0$, $P_{33} = = 1$, $P_{37} = = 0$, and $P_{45} = = 1$. The stimuli possessing property combination 1 are S_1, S_6, S_7, S_8, and S_{10}, and have higher average scores on usage intention than the others in the 12 stimuli, where only S_1 possesses the best property combination. That is why S_1 has the best average score on usage intention. Stimulus S_4 performs the worst on usage intention because it possesses the non-recommended property combination 2. Stimulus S_4 is hence suggested to be redesigned.

Table 13
Importance evaluation of perceptual vocabulary words (resp. service property states) on usage intention (resp. perceptual vocabulary words).

Importance of P_j to K_t	Perceptual vocabulary word					
Service property state	K_2	K_{16}	K_{26}	K_{30}	K_{31}	K_{37}
P_7	0.75					
P_8	0.25					
P_{32}		1.0				
P_{33}			0.22		0.24	0.22
P_{37}			0.78	0.79	0.76	0.78
P_{45}				0.21		
Importance of K_t to UI	**0.09**	**0.13**	**0.07**	**0.19**	**0.12**	**0.40**
	Usage intention (UI)					

Note: Bold values represent the importance evaluation of perceptual vocabulary words to usage intention.

The average scores of usage intention of stimuli S_2, S_3, S_5, S_9, S_{11}, and S_{12} are more than the that of S_4 and lower than those of S_1, S_6, S_7, S_8, and S_{10}. In particular, the average scores of usage intention of S_2, S_3, and S_5 are lower than 2. Such situations may result from S_2 and S_3 performing poorly in K_{30} and K_{31}, and from S_5 performing poorly in K_{16}, K_{30}, and K_{31}. Since stimuli S_2, S_3, and S_9 possess property combination 6, they can be improved by setting $P_{37} = = 0$ and $P_{33} = = 1$. In the same way, stimuli S_5 and S_{12} possessing property combination 8 can also be improved by setting $P_{37} = = 0$ and $P_{33} = = 1$. Stimulus S_{11} possesses property combination 9, and thus it can be improved by setting $P_{32} = = 0$.

4.2.5. Significance analysis of demographic and usage behavior variables on the important perceptual vocabularies

D2DD service providers may be interested in that the important perceptual responses are from which service users so that they can focus on their target markets. Accordingly, based on a sample of 298 responses, one-way analysis of variance (ANOVA) is implemented to find out the significance of demographic and usage behavior variables on the important perceptual vocabulary words including K_2, K_{16}, K_{26}, K_{30}, K_{31}, and K_{37}. The analytical results are given in Table 19. At a significance level 0.05, both gender and usage frequency (i.e. how frequently have you used D2DD service in the past year) variables are found to have significant relationship with K_{30} (i.e. delivery service is high quality) and K_{31} (i.e. delivery service is convenient). The other variables have no significant relationship with any important perceptual vocabulary words. The comparison between the mean score of male with the one of female on K_{30} (resp. K_{31}) shows that the females are more satisfied with the quality (resp. convenience) of delivery service. The service providers can highlight the high-quality and convenient delivery service when they focus on the female market. Similarly, the users whose usage frequencies exceed 6 times in the past year are more satisfied with the quality and convenience of delivery service. That is, the users may not feel high-quality and convenient until they have more than 6 times of usage frequencies in the past year.

5. Conclusions and managerial implications

Since Kano et al. (1984) stressed that customers usually expect to get great 'vibes' when using services, this paper develops an approach integrating Kansei Engineering and a decision tree for service design. Different service providers can innovate or improve their services and obtain marketing applications through the proposed approach. That is, the proposed approach considering the customers' perceptions in the process of service design is valuable. In order to demonstrate the performance and usability of the approach, a D2DD service is taken as an

Table 14
Recommendation of D2DD design for rule 1.

D2DD combination No.	D2DD property	Recommended property state
Property combination 1	E_{12}: Package tracking service	$P_{32} = = 0$: shipper provides the package tracking service
	E_{13}: Designate the delivery location	$P_{33} = = 1$: shipper can designate the delivery address, the D2DD service provider's business locations, or its cooperative receiving locations
	E_{14}: Designate the arrival time	$P_{37} = = 0$: shipper can designate a time period of a day or a date

Table 15
Non-recommended D2DD design for rule 3.

D2DD combination No.	D2DD property	Non-recommended property state
Property combination 2	E_{14}: Designate the arrival time	$P_{37} = = 1$: shipper cannot designate a time period of a day and a date
	E_{18}: Cash on delivery	$P_{45} = = 1$: receiver can pay to the D2DD service provider upon receipt of the package or pay by using the billing service provided by the D2DD service provider

Table 16
Non-recommended D2DD design for rule 4.

D2DD combination No.	D2DD property	Non-recommended property state
Property combination 3	E_1: Way to use service	$P_7 = = 1$: shipper can only go to the package receiving locations of D2DD service providers in person
	E_2: Cooperative package receiving locations	$P_8 = = 1$: the cooperative package receiving locations include convenience stores or other locations
	E_{14}: Designate the arrival time	$P_{37} = = 1$: shipper cannot designate a time period of a day and a date
	E_{18}: Cash on delivery	$P_{45} = = 0$: receiver can only pay to the D2DD service provider upon receipt of the package, or no cash on delivery is allowed
Property combination 4	E_1: Way to use service	$P_7 = = 1$: shipper can only go to the package receiving locations of D2DD service providers in person
	E_2: Cooperative package receiving locations	$P_8 = = 1$: the cooperative package receiving locations include convenience stores or other locations
	E_{13}: Designate the delivery location	$P_{33} = = 0$: shipper can only designate the delivery address or the D2DD service provider's business locations
	E_{14}: Designate the arrival time	$P_{37} = = 0$: shipper can designate a time period of a day or a date

Table 17
D2DD design for rule 6.

D2DD combination No.	D2DD property	Corresponding property state
Property combination 5	E_1: Way to use service	$P_7 = = 1$: shipper can only go to the package receiving locations of D2DD service providers in person
	E_2: Cooperative package receiving locations	$P_8 = = 0$: the D2DD service provider only has convenience stores; there is no cooperative package receiving location
	E_{14}: Designate the arrival time	$P_{37} = = 1$: shipper cannot designate a time period of a day and a date
	E_{18}: Cash on delivery	$P_{45} = = 1$: receiver can pay to the D2DD service provider upon receipt of the package or pay by using the billing service provided by the D2DD service provider
Property combination 6	E_1: Way to use service	$P_7 = = 0$: shipper can not only go to the package receiving locations of D2DD service providers in person but also has other ways to use the pick-up service
	E_{14}: Designate the arrival time	$P_{37} = = 1$: shipper cannot designate a time period of a day and a date
	E_{18}: Cash on delivery	$P_{45} = = 0$: receiver can only pay to the D2DD service provider upon receipt of the package, or no cash on delivery is allowed
Property combination 7	E_1: Way to use service	$P_7 = = 1$: shipper can only go to the package receiving locations of D2DD service providers in person
	E_2: Cooperative package receiving locations	$P_8 = = 0$: the D2DD service provider only has convenience stores; there is no cooperative package receiving location
	E_{13}: Designate the delivery location	$P_{33} = = 0$: shipper can only designate the delivery address or the D2DD service provider's business locations
	E_{14}: Designate the arrival time	$P_{37} = = 0$: shipper can designate a time period of a day or a date
Property combination 8	E_1: Way to use service	$P_7 = = 0$: shipper can not only go to the package receiving locations of D2DD service providers in person but also has other ways to use the pick-up service
	E_{13}: Designate the delivery location	$P_{33} = = 0$: shipper can only designate the delivery address or the D2DD service provider's business locations
	E_{14}: Designate the arrival time	$P_{37} = = 0$: shipper can designate a time period of a day or a date

Table 18
D2DD design for rule 7.

D2DD combination No.	D2DD property	Corresponding property state
Property combination 9	E_{12}: Package tracking service	$P_{32} = = 1$: no tracking service
	E_{13}: Designate the delivery location	$P_{33} = = 1$: shipper can designate the delivery address, the D2DD service provider's business locations, or its cooperative receiving locations
	E_{14}: Designate the arrival time	$P_{37} = = 0$: shipper can designate a time period of a day or a date

Table 19
Analysis of demographic and usage behavior variables associated with important perceptual vocabularies by one-way ANOVA.

		Important perceptual vocabulary																	
		K_{37}			K_{30}			K_{16}			K_{31}			K_{2}			K_{26}		
Variable		M	F	Sig.	M	F	Sig.	M	F	Sig.	M	F	Sig.	M	F	Sig.	M	F	Sig.
Gender	Male	2.03	1.871	0.172	2.02	4.649	**0.032***	2.29	1.184	0.277	2.01	4.830	**0.029***	2.14	1.275	0.190	2.23	1.616	0.205
	Female	2.16			2.23			2.4			2.23			2.27			2.35		
Age	< 20	2.10	1.073	0.370	2.30	2.290	0.060	2.35	0.962	0.429	2.25	2.027	0.091	2.40	0.380	0.823	2.40	1.189	0.316
	20–30	2.09			2.13			2.41			2.11			2.19			2.35		
	31–40	2.25			2.28			2.34			2.31			2.25			2.25		
	41–50	1.83			1.67			2.06			1.72			2.28			2.00		
	≥51	2.25			2.25			2.00			2.25			2.25			2.00		
Education level	High school or below	2.38	0.874	0.418	2.44	1.533	0.218	2.56	0.629	0.534	2.50	1.827	0.163	2.50	1.482	0.229	2.69	2.149	0.118
	College	2.09			2.10			2.33			2.10			2.24			2.27		
	Graduate school or above	2.10			2.19			2.39			2.17			2.14			2.31		
Occupation	Business	2.21	0.443	0.723	2.18	0.596	0.618	2.39	0.725	0.538	2.24	0.699	0.554	2.36	1.484	0.219	2.35	1.072	0.361
	Manufacturing	2.08			2.03			2.28			2.00			2.08			2.13		
	Public servants	2.08			2.29			2.17			2.13			2.33			2.21		
	Unemployed	2.08			2.14			2.40			2.14			2.18			2.35		
Monthly income	≤20,000	2.13	0.427	0.829	2.13	0.892	0.487	2.38	1.346	0.245	2.14	0.400	0.848	2.23	0.476	0.794	2.37	1.335	0.249
	20,001–30,000	1.96			2.15			2.50			2.12			2.12			2.46		
	30,001–40,000	2.05			2.02			2.17			2.02			2.19			2.07		
	40,001–50,000	2.20			2.34			2.55			2.27			2.30			2.32		
	50,001–60,000	2.00			2.29			2.29			2.14			2.00			2.14		
	≥60,001	2.18			2.04			2.18			2.18			2.32			2.25		
How long have you used D2DD service?	< 1 year	2.09		0.299	0.826	0.393	0.758	2.43	0.273	0.845	2.20	0.300	0.825	2.22	2.552	0.056	2.24	1.899	0.130
	1–2 years	2.05			2.16			2.38			2.20			2.39			2.43		
	3–4 years	2.17			2.07			2.30			2.10			2.05			2.37		
	≥5 years	2.11			2.20			2.37			2.11			2.25			2.17		
How frequently have you used D2DD service in the past year?	< 1 time	2.06	1.987	0.116	2.15	4.652	**0.003***	2.24	0.511	0.675	2.06	3.358	**0.019***	2.22	0.407	0.748	2.19	1.304	0.273
	2–3 times	2.06			2.13			2.38			2.18			2.21			2.30		
	4–6 times	2.02			1.84			2.43			1.86			2.14			2.27		
	≥7 times	2.34			2.43			2.38			2.36			2.31			2.47		
What kind of merchandise do you usually send?	Book/ documents	2.04	0.210	0.889	1.99	1.821	0.143	2.37	0.278	0.841	2.01	1.680	0.171	2.16	0.226	0.878	2.38	0.520	0.669
	Daily commodities	2.14			2.19			2.34			2.15			2.26			2.26		
	Gifts	2.13			2.29			2.45			2.34			2.21			2.36		
	Other	2.10			2.03			2.30			2.07			2.20			2.27		

Note: M = Mean; Sig. = Significance.
* Significant at p-value < 0.05.

example. This study explores 27 D2DD properties, 65 property states, and 38 perceptual vocabulary words, and then analyzes the relationship between the properties and perceptual vocabulary for D2DD design and improvement. By comparing the decision tree with the logistic regression in the analysis stage of Kansei Engineering, the decision tree is demonstrated to have better accuracy and be easier interpreted than the logistic regression. The obtained decision tree models show that several D2DD properties affect user experience as reflected in several crucial perceptual vocabulary words. Hence, through this questionnaire, we verify that the crucial D2DD properties utilized in the twelve stimuli affect these perceptive variables effectively.

The findings of this study reveal a significant relationship between the space of six property states (i.e. P_7, P_8, P_{32}, P_{33}, P_{37}, and P_{45}) and the Kansei semantic space (i.e. K_2, K_{16}, K_{26}, K_{30} K_{31}, and K_{37}) in terms of usage intention. These crucial perceptual vocabulary words are ranked according to their importance in constructing the decision tree model, and show that most of the respondents are concerned about the package delivery service. The addressed creative property states are not included in the six property states. However, the property state P_{32} involves whether to provide a tracking service, and thus is suggested to be "$P_{32} = = 0$", i.e. providing the tracking service. In other words, "$P_{32} = = 0$" means that D2DD service providers can consider the creative service, i.e. P_{30}: shipper can undertake package tracking through telephone, computer, or mobile app, as part of their service innovation. Furthermore, the analytical results are classified into three scenarios based on positive, negative, and neutral usage intention, as discussed below.

i. The first scenario describes which crucial perceptual responses result in positive usage intention and which important property states result in these crucial perceptual responses. The crucial perceptual responses are $K_{16} \geq 2$ and $K_{37} = = 3$, and the crucial property states are $P_{32} = = 0$, $P_{33} = = 1$ and $P_{37} = = 0$.
ii. The second scenario describes which crucial perceptual responses result in negative usage intention and which important property states result in these crucial perceptual responses. For the crucial perceptual responses and the non-recommended property states, refer to Tables 10, 12, 14, and 15.
iii. The third scenario describes which crucial perceptual responses result in the neutral response to the usage intention question and which important property states result in these crucial perceptual responses. For the crucial perceptual responses and the corresponding property states, refer to Tables 10, 12, 16 and 17.

D2DD companies can survey their current services according to the above three scenarios. If their D2DD fits the first scenario, it is suggested that the companies keep the services unchanged, or enhance their D2DD through creative services and even evoke customers' 'good vibes' to acquire the customers' loyalty. If their D2DD fits the second scenarios, it is suggested that the companies improve or redesign their services.

6. Limitations and future research

Aside from the academic and practical contributions presented in this study, there exist several limitations to be discussed in future work, as follows:

(1) The analytical results may not be fully suitable for use in other countries, since the respondents and D2DD service providers are all in Taiwan. Therefore, research findings should be further verified in different countries.
(2) This research domain is limited to the C2C and ambient-temperature D2DD, and this study only considers shippers as D2DD users. Such limitations may omit some perceptual variables. Future research could amplify the scope of collected perceptual vocabulary words from different users such as the users of C2C and cold chain D2DD, users of B2C and ambient-temperature D2DD, and users of B2C and cold chain D2DD.
(3) Regarding the form of stimulus, if allowed by time and economic resources, we recommend the presentation of stimuli through audio or images. In that way, the content of stimuli can be displayed in a more realistic manner, and the research results may better tally with users' actual experiences.

In addition to the above future works, it is considerable to integrate Kansei Engineering and QFD for service design and improvement. One of main input of QFD is a list of specific customer needs that are usually obtained according to Voice of Customer (VOC) (Franceschini & Maisano, 2015). Nevertheless, translating VOC into specific customer needs is a difficult task, especially for sensory products or services. Accordingly, Kansei Engineering can be utilized to identify the customer needs by analyzing VOC.

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